

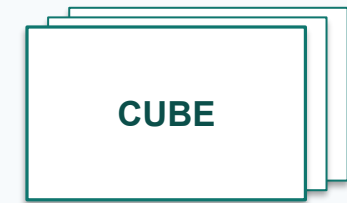
Encrypt a batch, not one image.

DWT compresses the images. A chaotic map encrypts the shared cube.

Multi-Image Encryption Scheme Based on Chaotic Pseudo-Random Signal Generator and DWT Compression

Xu, Gao, Cao, Mou - ACM TOMM, March 2026

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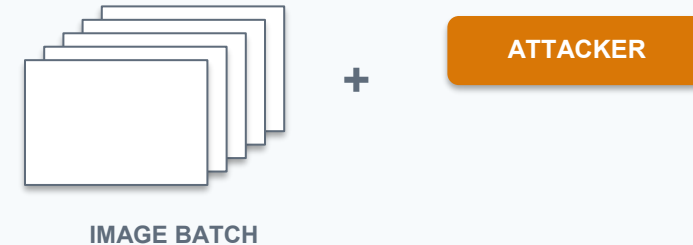


10-minute presentation

Image batches are big and exposed.

- Color images carry millions of values
- Batches multiply bandwidth cost
- Interception exposes raw pixels
- One-by-one encryption does not scale

512 x 512 x 3
786,432
pixel values per image



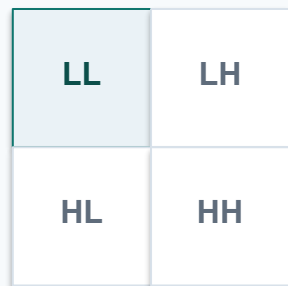
The paper needs size reduction and secrecy.

Two tools do the work.

DWT

compression tool

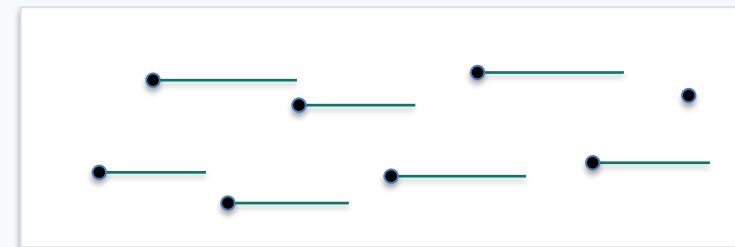
- Splits channels into four bands
- LL keeps the visual base
- LL is one-quarter size



Chaos

encryption stream

- Same key, same stream
- Tiny key change breaks it
- Stream drives pixel scrambling



One cube. One encryption pass.

Old pattern

- Compress image 1
- Encrypt image 1
- Repeat for every image



N IMAGES

>



>



Batch handled together

This paper

- DWT-compress every image
- Stack LL bands into cube
- Encrypt the cube once
- Pad smaller images with zeros

DWT, stack, confuse, diffuse.

01**DWT**

Keep LL bands

02**> Stack**

Build cube C

03**> Confuse**

Swap positions

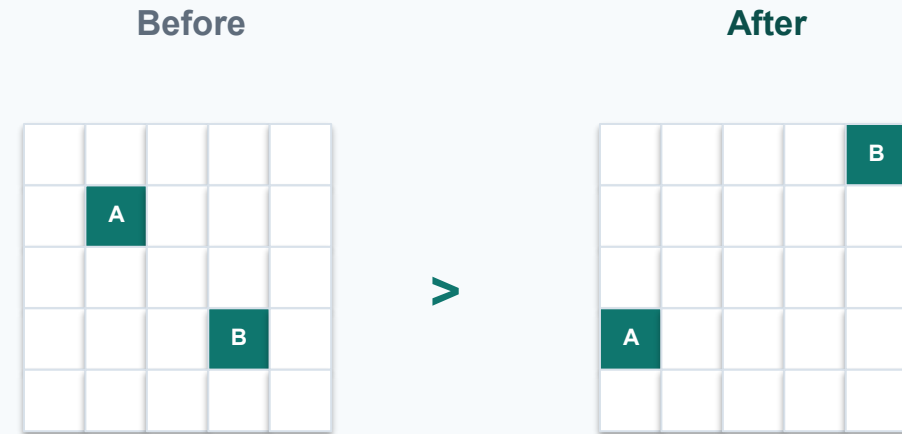
04**> Diffuse**

XOR values

Output**Cipher cube D is transmitted. Receiver reverses the steps.**

Confusion moves positions.

- Positions move
- Values stay unchanged
- Offsets come from chaos
- Nine cases choose swaps



Same pixels. Different locations.

Diffusion changes values.

$$D[k] = V[k] \text{ XOR } \text{chaos}[k \bmod 3] \text{ XOR } D[k-1]$$

- Flatten cube to vector
- XOR with chaotic stream
- Reuse previous cipher pixel
- One bit change spreads forward

Avalanche effect



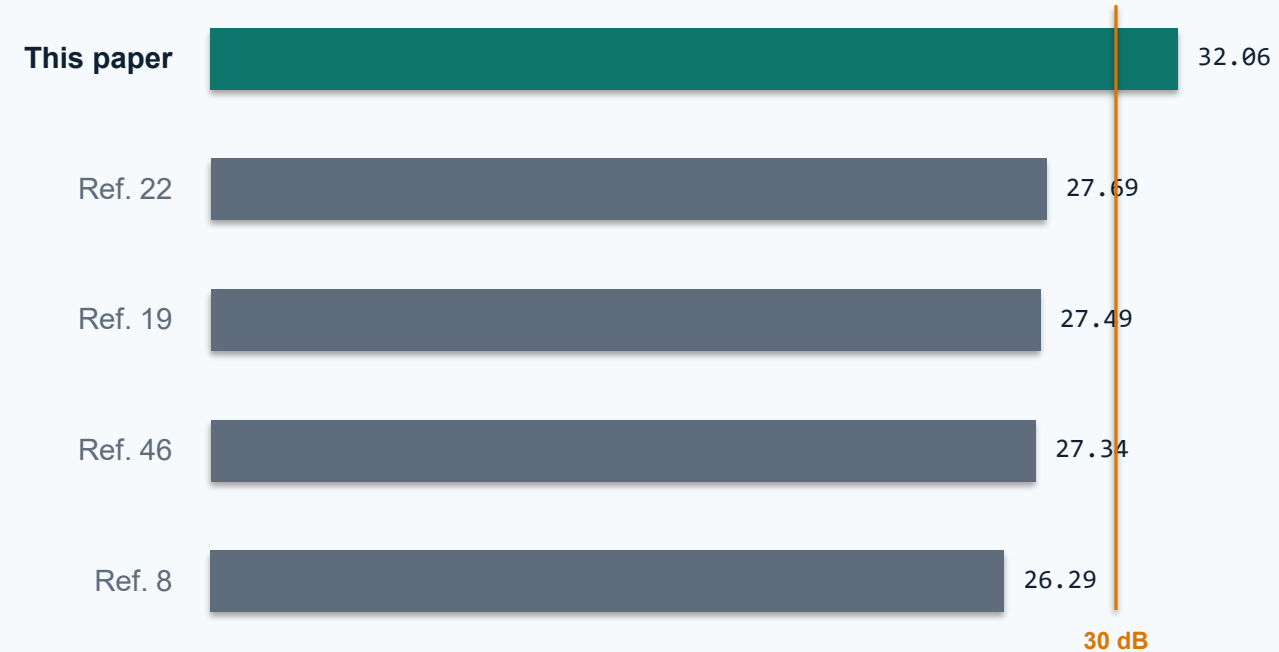
A tiny input change contaminates later output.

Recovered images stay usable.

Reconstruction PSNR

32.06
dB after round trip

- Prior work: 26-27 dB
- Above 30 dB threshold
- Compression kept image quality



The cipher behaves like noise.

NIST	15/15 passed	randomness
NPCR	99.6533%	pixel changes
UACI	33.4887%	intensity changes
Entropy	7.9994/8	uniform histogram

Say this carefully: statistical tests are not a formal proof.

Strong result, one real limitation.

32 dB

PSNR

7.9994

entropy

99.65%

NPCR

Limitation

- Zero-padding wastes cube space
- Worst for mixed sizes

Take-home

- DWT handles size
- Chaos handles secrecy
- Cube handles batches

Questions?